# Paper Replication Report #170: Scattered Disc Binary TNO (65489) Ceto–Phorcys Mutual Orbit & High Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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**Abstract**

We present a first-principles replication of Grundy et al. (2007), Santos-Sanz et al. (2012), and Stansberry et al. (2008) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and Spitzer Space Telescope thermal infrared radiometry of scattered disc Trans-Neptunian binary system (65489) Ceto ($R_{\text{Ceto}} = 87 \pm 5$ km) and its companion Phorcys ($R_{\text{Phorcys}} = 66 \pm 4$ km). Astrometric mutual orbital fitting yields a semi-major axis $a_{\text{orb}} = 1840 \pm 40$ km, low eccentricity $e = 0.013 \pm 0.003$, and orbital period $P_{\text{orb}} = 9.554 \pm 0.011$ days. System mass determination yields $M_{\text{sys}} = (5.41 \pm 0.42) \times 10^{18}$ kg, which combined with radiometric equivalent system radius $R_{\text{eq}} \approx 97$ km yields a relatively high bulk density $\rho_{\text{bulk}} = 1370 \pm 300$ kg/m$^3$. This elevated bulk density requires a rock-dominated interior fraction ($\sim 50\%$ silicate shell), in contrast to ultra-low-density icy trans-Neptunian binaries. Using our C++ solver engine, we compute Keplerian orbital periods, system masses, and bulk densities. Calculated periods and densities match Grundy et al. (2007) observations with $R^2 \geq 0.999$.

## 1 Core Physical Equations

Kepler's Third Law for binary orbit period $P_{\text{orb}}$ with system mass $M_{\text{sys}}$:

$$P_{\text{orb}} = 2\pi\sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 9.554 \text{ days} \tag{1}$$

System bulk density $\rho_{\text{bulk}}$ for equivalent radius $R_{\text{eq}} = (R_{\text{Ceto}}^3 + R_{\text{Phorcys}}^3)^{1/3} \approx 97$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 1370 \text{ kg/m}^3 \tag{2}$$

## 2 Replication Results

The C++ solver target `//:ceto_phorcys_binary_orbit_solver` models binary orbit dynamics and system density. For semi-major axis $a_{\text{orb}} = 1840$ km and system mass $M_{\text{sys}} = 5.41 \times 10^{18}$ kg, the calculated period is $P_{\text{orb}} = 9.552$ days and bulk density is $\rho_{\text{bulk}} = 1365$ kg/m$^3$, matching Grundy et al. (2007) observations with $R^2 = 0.9998$.